24 — Ammonia Main + MGO Aux: an 11× factor gap on one vessel

Proves · Ammonia main + MGO aux: an 11× factor gap on one vessel, the strongest fuel guard in the suite.

Mechanics this scenario turns on

- ME and AE each use **their own fuel's** CO2 factor (MGO/MDO 3.93267 · HFO 3.114 · LNG 2.753 · Ammonia 0.35154). The two per-engine cards must sum to the panel total.

Panels described below · Why this is the strongest fuel guard in the suite · Verified figures

Anything not described here — the mechanics above name the step that produced it; `00-ORIENTATION` Part 6 has the full number-to-step index.

Trust · characterisation snapshot, generated from the code. It detects change; it does NOT prove correctness. Figures marked *pending reference verification* have never been checked against anything outside the application.

Read after · scenario 13.

The widest CO2 spread the model can produce, on scenario 19's plant so both engines burn.

```
Ammonia  0.35154 kg CO2 / kg fuel
MGO      3.93267
ratio    11.2 ×
```

Why this is the strongest fuel guard in the suite

Epic 3 replaced a single `Co2Factor` with a per-engine lookup. A regression that quietly falls back to one constant would be invisible on every MDO/MGO scenario — the two factors are identical. It would also be nearly invisible on 13 (LNG/MGO, 2.753 vs 3.93267, a 30 % gap).

Here it is unmissable.

Verified figures

```
baseline FOC   8 861.6 t/yr   =  ME 6 880.9   +  AE 1 980.7
baseline CO2  10 208   t/yr   =  ME  2 419    +  AE  7 789
```

Hand-check

```
ME : 6 880.9 × 0.35154 =  2 419   ✓
AE : 1 980.7 × 3.93267 =  7 789   ✓
sum                    = 10 208   ✓
```

Note the inversion: the main engine burns **3.5× more fuel** than the aux side yet emits **3.2× less CO2**. A single-constant model cannot produce that shape at all — if this snapshot ever shows `baselineCO2 ≈ totalFOC × one factor`, the per-engine path has regressed.

Fuel price is Ammonia's default **$1 350/ton**, the most expensive in the table — so the cost panel and the CO2 panel move in opposite directions, which is the real-world trade this feature exists to show.

Takeaway: cheapest ≠ cleanest, and the model must keep the two axes independent. This scenario fails loudly if they are ever collapsed.
